Laboratory Assignment 5

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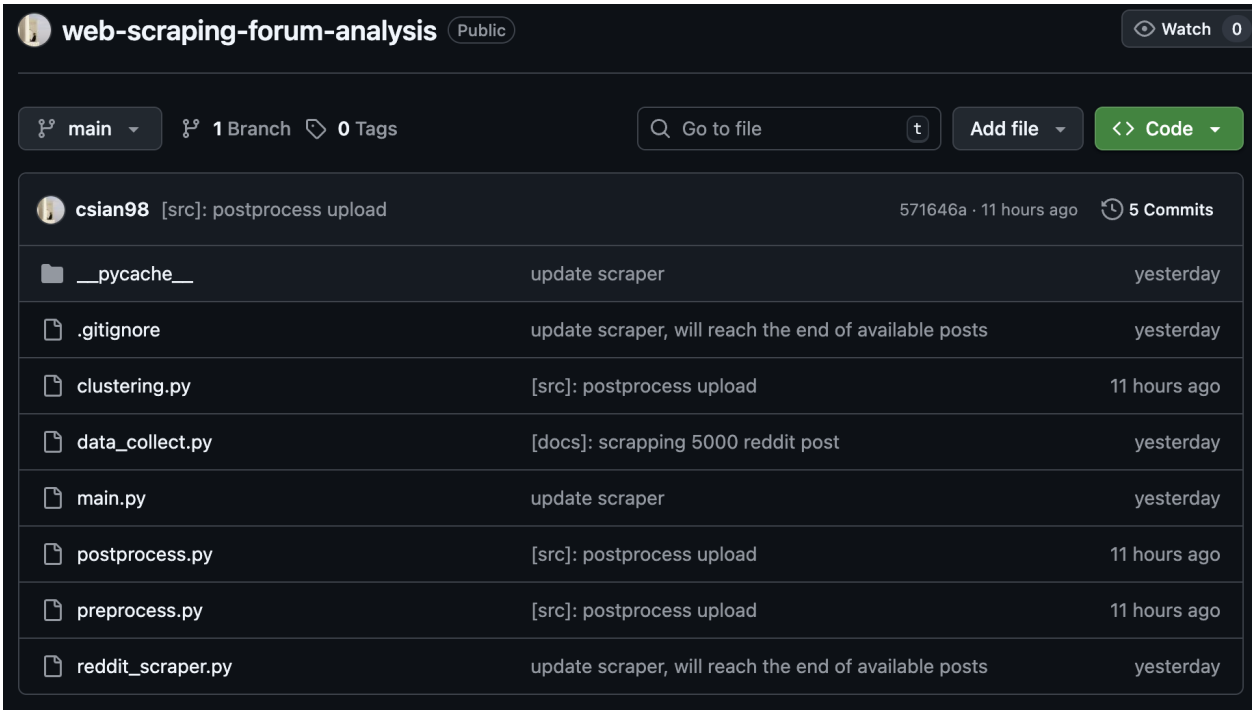


https://github.com/csian98/web-scraping-forum-analysis

Group meeting

2/11 WED 4:00 p.m - 4:30 p.m

First group meeting, setting up the overall outline and discussing data scraping from Reddit. We discussed two main methods: using the Reddit API and using Scrapper.
Assuming that here will be a delay in issuing keys for API use, we started developing a post scrapper using the static scraping (json) and dynamic scraping methods.

2/12 THU 11:00 a.m - 11:30 a.m

We decided to collect data using the scraper we completed the day before. We decided on the overall data collection format and completed the collaboration setup to automate the entire process with Snowflake.

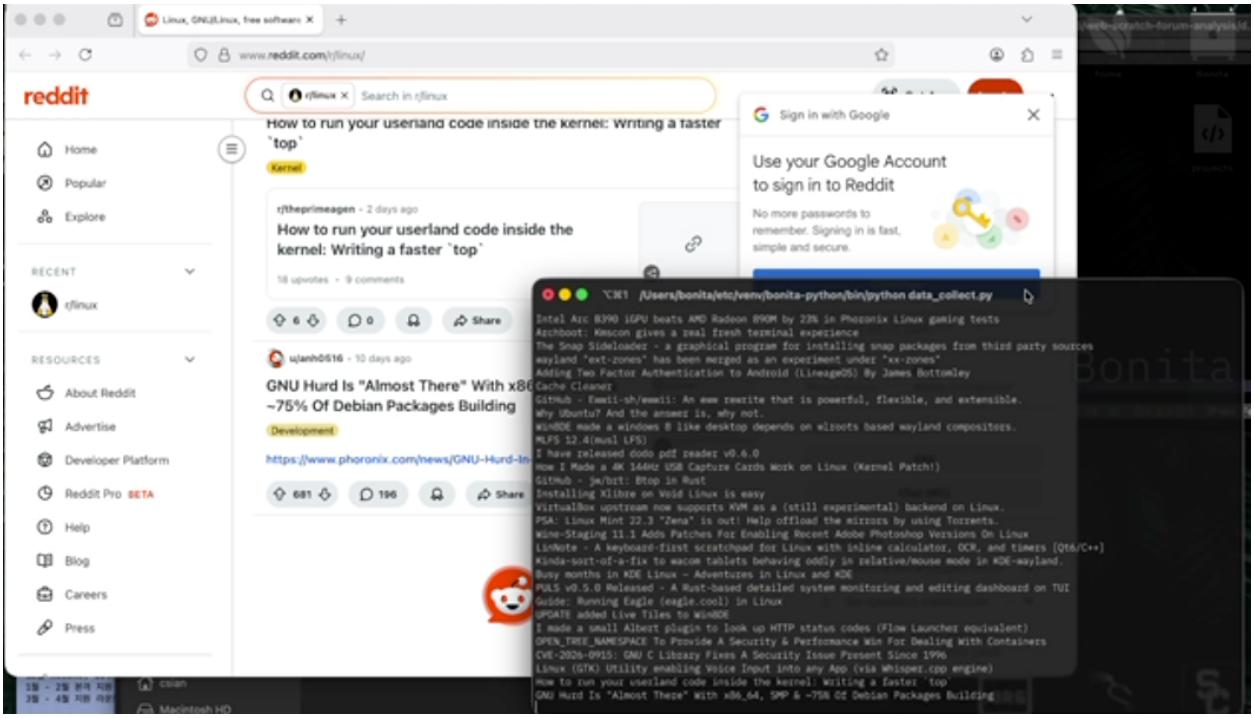
2/13 FRI 5:30 p.m - 6:30 p.m

Completed overall data collection (5,000 posts). Discussed data proprocessing and the preprocessing process for clustering analysis. We completed the overall data collection (5,000 posts). We discussed data preprocessing and the preprocessing process for clustering analysis. In

summary, after appropriate preprocessing, six subreddit data sets were stored in the SnowFlake cloud database. Embedding (doc2vec) were used to encapsulate the information, and clustering was then applied.

Data Scraping

We developed an application that scrape posts from Reddit's multiple subreddits, handle data preprocessing, apply clustering algorithms, store them in Snowflakes in vector format, and design automation process.



Our first attempt was to scrape posts from a single subreddit by repeatedly manipulating the mouse wheel using a dynamic scrapper (playwright). However, the number of posts that could be obtained for one topic using this method was approximately 500, and the processing speed was slower than that of the method using json used later.

```
~/synology-reliquia/academy/USC/MSDS/DSCI-560/laboratory/lab5/web-scratch-forum-analysis (main) » python main.py 500 --o
utput static-scrapping.txt --subreddit linux
======================================================================
REDDIT SCRAPER & PREPROCESSOR
Subreddit: r/linux
======================================================================

Fetching posts from Reddit...
Fetched 100/500 posts...
Fetched 200/500 posts...
Fetched 300/500 posts...
Fetched 400/500 posts...
```

The second method is to scrape the JSON format. We set the maximum posts to scrape per request and the delay time between each requests to respect the scraping policy.
Get the posts' information by scraping through the .json site. After each requests, get the following posts by loading the after parameter until there's no more posts available or it reach the number of posts to be fetched. Return the list of posts.

```
~/synology-reliquia/academy/USC/MSDS/DSCI-560/laboratory/lab5/web-scratch-forum-analysis/data (main) » ls -al
total 18008
drwxr-xr-x   6 csian  staff      192 Feb 13 20:58 .
drwxr-xr-x  17 csian  staff      544 Feb 13 21:04 ..
-rw-r--r--@  1 csian  staff     6148 Feb 13 20:23 .DS_Store
-rw-r--r--   1 csian  staff  3664367 Feb 12 20:49 dynamic-scrapping.txt
-rw-r--r--   1 csian  staff  4971906 Feb 13 20:58 raw_posts.csv
-rw-r--r--   1 csian  staff   569689 Feb 13 19:58 static-scrapping.txt
```

Preprocessing

After removing all advertisements and promoted posts, we extract some details from the json data. The process includes remove HTML tags and special characters, convert timestamps, mask usernames, and some basic data formatting. The processed posts are then returned for storage. We also considered a method of calculating and using the TF-IDF of each word token to extract the topic of the post, but it was difficult to use due to the relatively small number of post data and short post text.

```
~/synology-reliquia/academy/USC/MSDS/DSCI-560/laboratory/lab5/web-scratch-forum-analysis/data (main) » cat raw_posts.csv
 | head -n 2
post_id,title,selftext,author_hash,score,upvote_ratio,num_comments,created_utc,url,permalink,is_self,link_flair_text,dom
ain,subreddit,is_video,over_18
1qzv8yw,"Weekly Entering & Transitioning - Thread 09 Feb, 2026 - 16 Feb, 2026","Welcome to this week's entering & transi
tioning thread! This thread is for any questions about getting started, studying, or transitioning into the data science
 field. Topics include: * Learning resources (e.g. books, tutorials, videos) * Traditional education (e.g. schools, degr
ees, electives) * Alternative education (e.g. online courses, bootcamps) * Job search questions (e.g. resumes, applying,
 career prospects) * Elementary questions (e.g. where to start, what next) While you wait for answers from the community
, check out the [FAQ](https://www.reddit.com/r/datascience/wiki/frequently-asked-questions) and Resources pages on our w
iki. You can also search for answers in [past weekly threads](https://www.reddit.com/r/datascience/search?q=weekly%20thr
ead&restrict_sr=1&sort=new).",user_ac31a110,6,0.76,7,2026-02-08 21:01:37,https://www.reddit.com/r/datascience/comments/1
qzv8yw/weekly_entering_transitioning_thread_09_feb_2026/,https://www.reddit.com/r/datascience/comments/1qzv8yw/weekly_en
tering_transitioning_thread_09_feb_2026/,True,,self.datascience,datascience,False,False
```

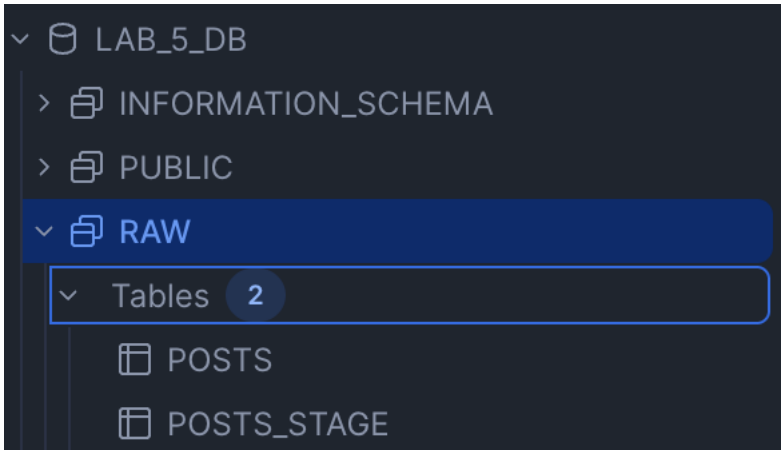
Vectorization (Embedding)

We use doc2vec method to turn text data into vector data. We do so in order to have further semantic and similarity comparisons, which is a part of our clustering methods.

```
X = (data['title'] + data['selftext']).to_list()
X = list(map(lambda x: post.tokenize(post.clean_text(x)), X))
X = post.doc2vec(X, 20, 2, 50)
```

Our group performed document embedding with 5, 10, and 20 dimensional data, respectively, and decided to embed the data in 20 dimensions, which showed the best performance when projected into 2 dimensions for visualization.

Storing Data on SnowFlake

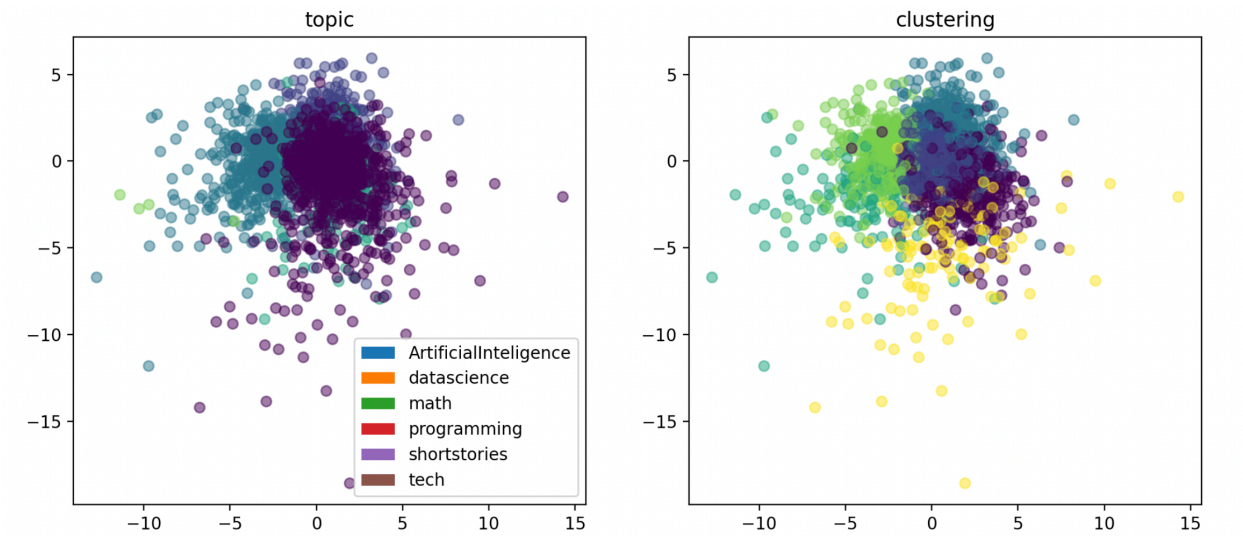


This is a snapshot of our Snowflakes database overview. We decided to use Snowflakes instead of traditional SQL database because it provides better concurrency control and this technology is getting more and more popular in today's job market.
The code in snowflake_util.py shows how we upload new post data into Snowflake staging and then merge it into the main posts table.

```
(venv) viraj@virajpc:~/Documents/web-scraping-forum-analysis$ python3 store_data.py 24
Total posts fetched: 4

Preprocessing posts from r/ArtificialInteligence...
Processed 4 posts (filtered out 0 promoted/ads)

Subreddit: r/shortstories
====================================================================

Fetching posts from Reddit...
Fetched 4/4 posts...
Total posts fetched: 4

Preprocessing posts from r/shortstories...
Processed 4 posts (filtered out 0 promoted/ads)

Fetch completed
Number of raw posts:24

Preprocessing completed.

Loading data into Snowflake staging table...
Successfully loaded 24 rows into Snowflake

Merging data from staging to main table...
Merge completed
Rows inserted: 2, Rows updated: 22
```

Afterwards, the entire code was organized and preprocessed to store the data in snow flake. Each post's topic was stored separately with k=3 using TF-IDF.

Clustering Analysis



For the final analysis, 5000 posts from six subreddits Artificial intelligence, Data Science, math, Programming, Short stories, and Tech -- were collected and stored in the SnowFlake database.

The entire data was embedded into a 20-dimensional vector using doc2vec, and clustering also used a 20-dimensional vector. For visualization, PCA was used to draw a graph using the two dimensions with the highest variance.

The AI and short stories topics appear to have formed relatively well-clustered data. However, other topics were not clearly distinguishable, likely due to their high similarity.

Afterwards, we attempted clustering k=2 by dividing the subtopics into tech and non-tech, excluding short stories, among all subtopics, and confirmed that the classification was performed with much higher accuracy than before.

Code Automation

For newly added Reddit posts, we'll store additional data, perform a new clustering fit, and then perform clustering on the input text. We're currently developing automation code to implement the above process.